

Contact Information

Department of Linguistics
Stony Brook University
Stony Brook, NY 11794-4376
USA

mail@thomasgraf.net
<https://thomasgraf.net>



Research Interests

- Mathematical linguistics
 - Subregularity in phonology, morphology, and syntax
 - Minimalist grammars
 - Model-theoretic syntax
- Parsing theory & syntactic processing
- Generative syntax
 - Binding
 - Economy, locality and encoding of constraints
 - Island constraints
 - Morphosyntax

Academic Employment and Affiliations

2019–present	Associate Professor Department of Linguistics, Stony Brook University
2015–present	Affiliate Professor Institute for Advanced Computational Science (IACS), Stony Brook University
2013–2019	Assistant Professor Department of Linguistics, Stony Brook University

Education

2010–2013	University of California, Los Angeles Ph.D. in Linguistics • Thesis: <i>Local and Transderivational Constraints in Syntax and Semantics</i> • Committee: Edward P. Stabler (chair), David Kaplan, Edward L. Keenan, Dominique Sportiche
2008–2010	University of California, Los Angeles M.A. in Linguistics • Thesis: <i>Logics of Phonological Reasoning</i> • Committee: Edward P. Stabler (chair), Robert Daland, Edward L. Keenan, Kie Zuraw
2003–2007	University of Vienna Mag.phil. (≈ M.A.) in Linguistics • Thesis: <i>From Vagueness to Disaster: On the Intricacies of Feature Checking</i> • Committee: Wolfgang U. Dressler (chair), Chris Schaner-Wolles

Author information Throughout the CV, co-authors who were Stony Brook students when the work was done are marked with SBU. Co-authors who were students at other institutions are marked with an asterisk.

Journal Papers

- [11] **under review**. A cognitive economy argument for probe horizons. *Linguistic Inquiry*.
- [10] **2025**. The dose makes the poison: Chesi's vision and subregular syntax. *Italian Journal of Linguistics* 37. 85–98. <https://doi.org/10.26346/1120-2726-241>.
- [9] **2022**. Subregular linguistics: bridging theoretical linguistics and formal grammar. *Theoretical Linguistics* 48. 145–184. <https://doi.org/10.1515/tl-2022-2037>. 
- [8] **2022**. Diving deeper into subregular syntax. *Theoretical Linguistics* 48. 245–278. <https://doi.org/10.1515/tl-2022-2043>. 
- [7] **2021**. The computational unity of Merge and Move. *Evolutionary Linguistic Theory* 3. 154–180. <https://doi.org/10.1075/elt.00032.gra>.
- [6] **2021** (with Robert Pasternak^{SBU}). Cyclic scope and processing difficulty in a Minimalist parser. *Glossa* 6. 1–34. <https://doi.org/10.5334/gjgl.1209>. 
- [5] **2019**. Monotonicity as an effective theory of morphosyntactic variation. *Journal of Language Modelling* 7. 3–47. <https://doi.org/10.15398/jlm.v7i2.211>. 
- [4] **2017**. A computational guide to the dichotomy of features and constraints. *Glossa* 2. 1–36. <https://doi.org/10.5334/gjgl.212>. 
- [3] **2017**. The power of locality domains in phonology. *Phonology* 34. 385–405. <https://doi.org/10.1017/S0952675717000197>.
- [2] **2017** (with James Monette^{SBU} and Chong Zhang^{SBU}). Relative clauses as a benchmark for Minimalist parsing. *Journal of Language Modelling* 5. 57–106. <https://doi.org/10.15398/jlm.v5i1.157>. 
- [1] **2007**. Agreement with hybrid nouns in Icelandic. *Snippets* 16. Available at <http://www.ledonline.it/snippets/>, 7–8.

Books

- [1] **in progress**. *The little book of language, math, and computation* (Textbooks in Language Sciences). Available at <http://lamaco.thomasgraf.net>. Language Science Press.

Invited Book Chapters

- [3] **to appear**. Minimalism and computational linguistics. In Kleanthes K. Grohman & Evelina Leivada (eds.), *Handbook of Minimalism*. Preprint available at <https://lingbuzz.net/lingbuzz/005855>. Cambridge, MA: Cambridge University Press.
- [2] **to appear**. Distributed morphology and computational linguistics. In Artemis Alexiadou, Alec Marantz, Isabel Oltra-Massuet & Ruth Kramer (eds.), *Handbook of Distributed Morphology*. Cambridge, MA: Cambridge University Press.

- [1] **2024** (with Alëna Aksënova, Jonathan Rawski, and Jeffrey Heinz). The computational power of harmonic forms. In Harry van der Hulst (ed.), *Handbook of vowel harmony*, 437–451. Oxford, UK: Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780198826804.013.34>.

Refereed Book Chapters

- [2] **2012**. Concealed reference-set computation: How syntax escapes the parser's clutches. In Anna Maria Di Sciullo (ed.), *Towards a biolinguistic understanding of grammar. Essays on interfaces*, 339–362. Amsterdam: John Benjamins.
- [1] **2010**. Formal parameters of phonology: From Government Phonology to SPE. In Thomas Icard & Reinhard Muskens (eds.), *Interfaces: Explorations in logic, language and computation*, vol. 6211 (Lectures Notes in Artificial Intelligence), 72–86. Berlin: Springer.

Paper-Reviewed Proceedings Papers (full papers peer-reviewed upon submission)

- [31] **2025** (with Kenneth Hanson^{SBU}). Automata for subregular syntax: syntax with strings attached. In *Proceedings of the Society for Computation in Linguistics (SCiL) 2025*. <https://doi.org/10.7275/scil.3184>. 
- [30] **2025** (with Logan Swanson^{SBU} and Kenneth Hanson^{SBU}). Are syntactic categories ISL-2 inferrable? A corpus study. In *Proceedings of the Society for Computation in Linguistics (SCiL) 2025*. <https://doi.org/10.7275/scil.3170>. 
- [29] **2024**. Morphologically simplex D-quantifiers are strictly 2-local. In *Proceedings of the Society for Computation in Linguistics (SCiL) 2024*, 32–42. <https://doi.org/10.7275/scil.2127>. 
- [28] **2023** (with Charles Torres*, Kenneth Hanson^{SBU}, and Connor Mayer). Modeling island effects with probabilistic tier-based strictly local grammars over trees. In *Proceedings of the Society for Computation in Linguistics (SCiL) 2023*, 155–164. <https://doi.org/10.7275/nz4q-6b09>. 
- [27] **2023**. Subregular tree transductions, movement, copies, traces, and the ban on improper movement. In *Proceedings of the Society for Computation in Linguistics (SCiL) 2023*, 289–299. <https://doi.org/10.7275/tk1n-q855>. 
- [26] **2022** (with Hossep Dolatian and Shiori Ikawa*). Trees probe deeper than strings: an argument from allomorphy. In *Proceedings of the 19th SIGMORPHON workshop on computational research in phonetics, phonology, and morphology*, 51–60. Seattle, Washington: Association for Computational Linguistics. <https://doi.org/10.18653/v1/2022.sigmorphon-1.6>. 
- [25] **2022**. Typological implications of tier-based strictly local movement. In *Proceedings of the Society for Computation in Linguistics (SCiL) 2022*, 184–193. <https://doi.org/10.7275/gb65-ht31>. 
- [24] **2021** (with Hossep Dolatian and Aniello De Santo). Recursive prosody is not finite-state. In *Proceedings of SIGMORPHON 2021*, 11–22. <https://doi.org/10.18653/v1/2021.sigmorphon-1.2>. 
- [23] **2021** (with Kalina Kostyszyn^{SBU}). Multiple wh-movement is not special: the subregular complexity of persistent features in Minimalist grammars. In *Proceedings of the Society for Computation in Linguistics (SCiL) 2021*, 275–285. <https://doi.org/10.7275/2dfv-v219>. 

- [22] **2020**. Curbing feature coding: Strictly local feature assignment. In *Proceedings of the Society for Computation in Linguistics (SCiL) 2020*, 362–371. <https://doi.org/10.7275/f7y5-xz32>. 
- [21] **2020** (with Nazila Shafiei^{SBU}). The subregular complexity of syntactic islands. In *Proceedings of the Society for Computation in Linguistics (SCiL) 2020*, 272–281. <https://doi.org/10.7275/scil.1202>. 
- [20] **2019** (with Aniello De Santo^{SBU}). Sensing tree automata as a model of syntactic dependencies. In *Proceedings of the 16th meeting on the mathematics of language*, 12–26. Toronto, Canada: Association for Computational Linguistics. 
- [19] **2019** (with Aniello De Santo^{SBU}). Structure sensitive tier projection: applications and formal properties. In Raffaella Bernardi, Gregory Kobele & Sylvain Pogodalla (eds.), *Formal grammar*, 35–50. Berlin, Heidelberg: Springer. https://doi.org/10.1007/978-3-662-59648-7_3.
- [18] **2019** (with Mai Ha Vu* and Nazila Shafiei^{SBU}). Case assignment in TSL syntax: a case study. In Gaja Jarosz, Max Nelson, Brendan O'Connor & Joe Pater (eds.), *Proceedings of the Society for Computation in Linguistics (SCiL) 2019*, 267–276. <https://doi.org/10.7275/sy wz-xw23>. 
- [17] **2019** (with Nazila Shafiei^{SBU}). C-command dependencies as TSL string constraints. In Gaja Jarosz, Max Nelson, Brendan O'Connor & Joe Pater (eds.), *Proceedings of the Society for Computation in Linguistics (SCiL) 2019*, 205–215. <https://doi.org/10.7275/4rrx-x488>. 
- [16] **2018** (with Connor Mayer*). Sanskrit n-retroflexion is input-output tier-based strictly local. In *Proceedings of SIGMORPHON 2018*, 151–160. 
- [15] **2018**. Grammar size and quantitative restrictions on movement. In *Proceedings of the Society for Computation in Linguistics (SCiL) 2018*, 23–33. <https://doi.org/10.7275/R5348HJ8>. 
- [14] **2017**. Graph transductions and typological gaps in morphological paradigms. In *Proceedings of the 15th meeting on the mathematics of language (MOL 2017)*, 114–126. 
- [13] **2016** (with Alëna Aksënova^{SBU} and Sedigheh Moradi^{SBU}). Morphotactics as tier-based strictly local dependencies. In *Proceedings of the 14th SIGMORPHON workshop on computational research in phonetics, phonology, and morphology*, 121–130. 
- [12] **2016** (with Alëna Aksënova^{SBU} and Aniello De Santo^{SBU}). A single movement normal form for Minimalist grammars. In Annie Foret, Glyn Morrill, Reinhard Muskens, Rainer Osswald & Sylvain Pogodalla (eds.), *Formal Grammar: 20th and 21st international conferences, FG 2015, Barcelona, Spain, August 2015, revised selected papers. FG 2016, Bozen, Italy, August 2016*, 200–215. Berlin, Heidelberg: Springer. https://doi.org/10.1007/978-3-662-53042-9_12.
- [11] **2016** (with Aniello De Santo^{SBU} and Alëna Aksënova^{SBU}). An alternate view on strong lexicalization in TAG. In *Proceedings of the 12th international workshop on Tree Adjoining Grammars and related formalisms (TAG+12)*, 93–102. Düsseldorf, Germany. 
- [10] **2015** (with Brigitta Fodor^{SBU}, James Monette^{SBU}, Gianpaul Rachiele^{SBU}, Aunika Warren^{SBU}, and Chong Zhang^{SBU}). A refined notion of memory usage for Minimalist parsing. In *Proceedings of the 14th meeting on the mathematics of language (MOL 2015)*, 1–14. Chicago, USA: Association for Computational Linguistics. 

- [9] **2014** (with Bradley Marcinek^{SBU}). Evaluating evaluation metrics for Minimalist parsing. In *Proceedings of the 2014 ACL workshop on cognitive modeling and computational linguistics (CMCL)*, 28–36.
- [8] **2014**. Late Merge as lowering movement in Minimalist grammars. In Nicholas Asher & Sergei Soloviev (eds.), *LACL 2014*, vol. 8535 (Lecture Notes in Computer Science), 107–121. Heidelberg: Springer. https://doi.org/10.1007/978-3-662-43742-1_9.
- [7] **2014**. Models of adjunction in Minimalist grammars. In Glynn Morrill, Reinhard Muskens, Rainer Osswald & Frank Richter (eds.), *Formal Grammar 2014*, vol. 8612 (Lecture Notes in Computer Science), 52–68. Heidelberg: Springer. https://doi.org/10.1007/978-3-662-44121-3_4.
- [6] **2012**. Locality and the complexity of Minimalist derivation tree languages. In Philippe de Groot & Mark-Jan Nederhof (eds.), *Formal Grammar 2010/2011*, vol. 7395 (Lecture Notes in Computer Science), 208–227. Heidelberg: Springer. https://doi.org/10.1007/978-3-642-32024-8_14.
- [5] **2012**. Movement-generalized Minimalist grammars. In Denis Béchet & Alexander J. Dikovsky (eds.), *LACL 2012*, vol. 7351 (Lecture Notes in Computer Science), 58–73. https://doi.org/10.1007/978-3-642-31262-5_4.
- [4] **2012**. Reference-set constraints as linear tree transductions via controlled optimality systems. In Philippe de Groote & Mark-Jan Nederhof (eds.), *Formal Grammar 2010/2011*, vol. 7395 (Lecture Notes in Computer Science), 97–113. Heidelberg: Springer. https://doi.org/10.1007/978-3-642-32024-8_7.
- [3] **2011**. Closure properties of Minimalist derivation tree languages. In Sylvain Pogodalla & Jean-Philippe Prost (eds.), *LACL 2011*, vol. 6736 (Lecture Notes in Artificial Intelligence), 96–111. Heidelberg: Springer. https://doi.org/10.1007/978-3-642-22221-4_7.
- [2] **2010**. Some interdefinability results for syntactic constraint classes. In Christian Ebert, Gerhard Jäger & Jens Michaelis (eds.), *The mathematics of language*, vol. 6149 (Lecture Notes in Computer Science), 72–87. Heidelberg: Springer.
- [1] **2009**. Towards a factorization of string-based phonology. In Thomas Icard (ed.), *Proceedings of the 14th student session of the European summer school for logic, language and information (ESSLI)*, 72–84.

Abstract-Reviewed Proceedings Papers (only an abstract was peer-reviewed upon submission)

- [10] **2020**. Monotonicity in syntax. In Dun Deng, Fenrong Liu, Mingming Liu & Dag Westerståhl (eds.), *Monotonicity in logic and language*, vol. 12564 (Lecture Notes in Computer Science), 35–53. Berlin, Heidelberg: Springer. https://doi.org/10.1007/978-3-662-62843-0_3.
- [9] **2020** (with Sabine Laszakovits^{*}). Diagnosing movement via the absence of c-command relations. In *University of Pennsylvania working papers in linguistics*, vol. 26, 135–144. Article 16.
- [8] **2019**. A subregular bound on the complexity of lexical quantifiers. In Julian J. Schlöder, Dean McHugh & Floris Roelofsen (eds.), *Proceedings of the 22nd Amsterdam Colloquium*, 455–464.

- [7] **2018.** Locality domains and phonological c-command over strings. In *Nels 48: proceedings of the forty-eighth annual meeting of the north east linguistic society*, vol. 1, 257–270. Amherst, MA: GLSA.
- [6] **2018.** Why movement comes for free once you have adjunction. In Daniel Edmiston, Marina Ermolaeva, Emre Hakgüder, Jackie Lai, Kathryn Montemurro, Brandon Rhodes, Amara Sankhagowit & Miachel Tabatowski (eds.), *Proceedings of CLS 53*, 117–136.
- [5] **2014.** Feature geometry and the Person Case Constraint: an algebraic link. In *Proceedings of CLS 50*. To appear.
- [4] **2013.** The syntactic algebra of adjuncts. In *Proceedings of CLS 49*. To appear.
- [3] **2012.** Tree adjunction as Minimalist lowering. In *Proceedings of the 11th international workshop on Tree Adjoining Grammars and related formalisms (TAG+11)*, 19–27. 
- [2] **2012** (with Natasha Abner). Is syntactic binding rational? In *Proceedings of the 11th international workshop on Tree Adjoining Grammars and related formalisms (TAG+11)*, 189–197.
- [1] **2010.** Comparing incomparable frameworks: a model theoretic approach to phonology. *University of Pennsylvania Working Papers in Linguistics* 16(2). Article 10. 

Non-Refereed Publications

- [6] **2017.** Derivations as representations: News from the computational frontier. In Clemens Mayr & Edwin Williams (eds.), *Festschrift for Martin Prinzhorn*, vol. 82 (Wiener Linguistische Gazette), 61–69.
- [5] **2014.** Beyond the apparent: Cognitive parallels between syntax and phonology. In Carson T. Schütze & Linnaea Stockall (eds.), *Connectedness: Papers by and for Sarah van Wagenen*, vol. 18 (UCLA Working Papers in Linguistics), 161–174. 
- [4] **2012.** An algebraic perspective on the Person Case Constraint. In Thomas Graf, Denis Paperno, Anna Szabolcsi & Jos Tellings (eds.), *Theories of everything. In honor of Ed Keenan*, vol. 17 (UCLA Working Papers in Linguistics), 85–90. 
- [3] **2010.** A tree transducer model of reference-set computation. *UCLA Working Papers in Linguistics* 15, Article 4. 1–53. 
- [2] **2010.** *Logics of phonological reasoning*. University of California, Los Angeles MA thesis. 
- [1] **2007.** *From vagueness to disaster: on the intricacies of feature checking*. University of Vienna MA thesis.

Edited Publications

- [3] **in prep** (with Aniello De Santo), eds. *New Trends in Syntactic Islands*. (Languages).
- [2] **2012** (with Denis Paperno, Anna Szabolcsi, and Jos Tellings), eds. *Theories of Everything. In Honor of Ed Keenan*. Vol. 17. (UCLA Working Papers in Linguistics). 
- [1] **2010**, ed. *Papers in Mathematical Linguistics 1*. Vol. 15. (UCLA Working Papers in Linguistics). 

Invited Talks

- [33] **2023.** *Linguistics and symbolic computation in a world of large language models.* Invited talk, December 20, Austrian Research Institute for Artificial Intelligence (OFAI), Vienna, Austria.
- [32] **2023.** *Formal and substantive universals across language modules.* Invited talk, January 31, Surrey Linguistics Circle, Surrey, United Kingdom.
- [31] **2022.** *Syntactic tiers: empirical applications and challenges.* Invited talk, November 11, NYU, New York, New York.
- [30] **2021.** *Monotonicity as a high-level principle of grammar.* Invited talk, July 16, University of Leipzig, Leipzig, Germany.
- [29] **2021.** *Monotonicity as a third factor in syntax and morphology.* Invited talk, May 7, MIT, Boston, Massachusetts.
- [28] **2021.** *A computational Minimalist program for syntax and sentence processing.* Invited talk, April 8, CUNY Graduate School, New York City, New York.
- [27] **2019.** *Subregular linguistics for linguists.* Invited talk, November 8, University of Massachusetts Amherst, Amherst, Massachusetts.
- [26] **2019.** *Working the tail: improving UX by optimizing low-frequency items.* Invited talk, July 8, Kasisto Inc, New York City, New York.
- [25] **2019.** *Features: more trouble than they're worth?* Invited talk, May 27, Workshop on *Thirty Million Theories of Syntactic Features*, University of Tromsø, Tromsø, Norway.
- [24] **2018.** *Subregular syntax: the what, how, and why.* Invited talk, December 1, SYNC, Stony Brook University, Stony Brook, NY.
- [23] **2018.** *The surprising simplicity of syntax: derivation trees, subregular complexity, and what it implies for language and cognition.* Invited talk, November 26, Integrated Language Science and Technology Seminar, University of Pennsylvania, Philadelphia, PA.
- [22] **2018.** *Pogo sticks and wasps: a skeptic's guide to computational linguistics.* Invited talk, October 11, Kasisto Inc, New York City, New York.
- [21] **2017.** *One reason to move, a million reasons to be an island: Third-factor explanations from computational syntax.* Invited talk, December 15, Rutgers University, New Brunswick, NJ.
- [20] **2017.** *Subregular morpho-semantics: The expressive limits of monomorphemic quantifiers.* Invited talk, December 15, Rutgers University, New Brunswick, NJ.
- [19] **2017.** *Do we need features for morphosyntax?* Invited talk, June 26, Workshop on Morphomathematics, Leibniz-Zentrum für Allgemeine Sprachwissenschaft, Berlin, Germany.
- [18] **2017.** *It's a (sub-)regular conspiracy: Locality and computation in phonology, morphology, syntax, and semantics.* Invited talk, May 25–27, CLS, University of Chicago, Chicago, IL.
- [17] **2017.** *Fragments of first-order logic for linguistic structures.* Invited talk, April 21, UConn Logic Group, University of Connecticut, Storrs, CT.

- [16] **2016.** *Computational parallels across language modules*. Invited talk, September 12, Department of Linguistics, Yale University, New Haven, CT.
- [15] **2016.** *Computational lessons from and for language*. Invited talk, April 28, IACS, Stony Brook University, Stony Brook, NY.
- [14] **2015.** *Towards an algebraic morphosyntax*. Invited talk, December 16, Department of Theoretical and Applied Linguistics, Moscow State University, Moscow, Russia.
- [13] **2015.** *Computational unity across language modules*. Invited talk, December 15, Department of Theoretical and Applied Linguistics, Moscow State University, Moscow, Russia.
- [12] **2015.** *Adjuncts, conjuncts, o-juncts: Deriving strong island constraints*. Invited talk, Institute of Linguistics, Russian State University for the Humanities (RGGU), Moscow, Russia.
- [11] **2015.** *Syntax and phonology: A computational common core*. Department of Mathematical Linguistics, Saint Petersburg State University, St. Petersburg, Russia.
- [10] **2015.** *Formal processing theory, or parsing without parsers*. Slides of a talk given at the Workshop on Minimalist Parsing, October 9–12, MIT, Boston, MA.
- [9] **2014.** *Adjuncts, islands, algebra (with a sprinkling of semantics)*. Invited talk, November 4, CUNY Graduate Center, New York, NY.
- [8] **2014.** *Person case constraints and feature complexity in syntax*. Invited talk, January 9, Institut für Sprachwissenschaft, Universität Wien, Vienna, Austria.
- [7] **2013.** *Computational computational linguistics*. Invited talk, December 4, Department of Psychology, Stony Brook University, Stony Brook, NY.
- [6] **2013.** *Optionality implies islandhood*. Invited talk, November 1, Department of Linguistics, NYU, New York, NY.
- [5] **2013.** *Islands without islands*. Invited talk, October 4, Department of Linguistics and Cognitive Science, University of Delaware, Newark, DE.
- [4] **2013.** *Constraints emerge from Merge*. Invited talk, February 12, Department of Linguistics, Stony Brook University, Stony Brook, NY.
- [3] **2010.** *Syntax, semantics, pragmatics: Where do we find optimality?* Invited talk, December 15, Institut für Anglistik, Universität Göttingen, Göttingen, Germany.
- [2] **2010.** *Optimality conditions could care less about optimality*. Invited talk, December 10, Zentrum für Allgemeine Sprachwissenschaft (ZAS), Berlin, Germany.
- [1] **2010.** *Reference-set computation = Minimalism + transformational rules?* Invited talk, December 8, Institut für Linguistik, Universität Bielefeld, Bielefeld, Germany.

Invited Lectures, Mini-Courses & Summer Schools

- [3] **2023.** *Syntactic tiers for movement and agreement*. Mini-course, December 6–8, CRISSP KU Leuven, Brussels, Belgium.

- [2] **2021.** *The computational nature of language*. Summer course, July 26–30, ESSLLI, online.
- [1] **2021.** *Subregular linguistics for... well, linguists*. Mini-course, May 5–6, MIT, Boston, Massachusetts.

Other Talks & Posters

- [59] **2024.** *Morphologically simplex D-quantifiers are strictly 2-local*. Slides of a talk given at SCiL 2024, June 27–29, UC Irvine, Irvine, California.
- [58] **2023.** *Subregular tree transductions, movement, copies, traces, and the ban on improper movement*. Slides of a talk given at SCiL 2023, June 15–17, University of Massachusetts Amherst, Amherst, Massachusetts.
- [57] **2023** (with Charles Torres^{*}, Kenneth Hanson^{SBU}, and Connor Mayer). Modeling island effects with probabilistic tier-based strictly local grammars over trees. In Slides of a talk given at SCiL 2023, June 15–17, University of Massachusetts Amherst, Amherst, Massachusetts.
- [56] **2022.** *Typological implications of tier-based strictly local movement*. Slides of a talk given at SCiL 2022, February 7–9, online.
- [55] **2021** (with Hossep Dolatian and Aniello De Santo). *Recursive prosody is not finite-state*. Slides of a talk given at SIGMORPHON 2021, August 5, online.
- [54] **2021** (with Kalina Kostyszyn^{SBU}). *Multiple wh-movement is not special: the subregular complexity of persistent features in Minimalist grammars*. Slides of a talk given at SCiL 2021, February 15–19, online.
- [53] **2020.** *Monotonicity in syntax*. Slides of a talk given at the Second Tsinghua Interdisciplinary Workshop on Logic, Language, and Meaning (TLLM 2020), December 17–20, Beijing, China.
- [52] **2020.** *Curbing feature coding: strictly local feature assignment*. Slides of a talk given at SCiL 2020, January 2–5, co-located with the LSA 2020.
- [51] **2020** (with Nazila Shafiei^{SBU}). *The subregular complexity of syntactic islands*. Poster presented at SCiL 2020, January 2–5, co-located with the LSA 2020.
- [50] **2019.** *A subregular bound on the complexity of lexical quantifiers*. Slides of a talk given at the Amsterdam Colloquium, December 18–20, University of Amsterdam, Amsterdam, Netherlands.
- [49] **2019** (with Aniello De Santo^{SBU}). *Structure sensitive tier projection: applications and formal properties*. Slides of a talk given at the 24th Conference on Formal Grammar, August 11, Riga, Latvia.
- [48] **2019** (with Aniello De Santo^{SBU}). *Sensing tree automata as a model of syntactic dependencies*. Slides of a talk given at the 16th meeting on the mathematics of language (MOL 2019), July 18–19, University of Toronto, Toronto, Canada.
- [47] **2019** (with Sabine Laszakovits^{*}). *Diagnosing movement via the absence of c-command relations*. Slides presented at the 43rd Penn Linguistics Colloquium (PLC43), March 22–24, University of Pennsylvania, Philadelphia, PA.

- [46] **2019** (with Nazila Shafiei^{SBU}). *C-command dependencies as TSL string constraints*. Poster presented at *SCiL 2019*, January 3–6, co-located with the LSA 2019.
- [45] **2019** (with Sedigheh Moradi^{SBU} and Alëna Aksënova^{SBU}). *The computational cost of generalizations: an example from micromorphology*. Poster presented at *SCiL 2019*, January 3–6, co-located with the LSA 2019.
- [44] **2019** (with Mai Ha Vu^{*} and Nazila Shafiei^{SBU}). *Case assignment in TSL syntax: a case study*. Poster presented at *SCiL 2019*, January 3–6, co-located with the LSA 2019.
- [43] **2018** (with Connor Mayer^{*}). *Sanskrit n-retroflexion is input-output tier-based strictly local*. Poster presented at *SIGMORPHON 2018*, October 31, Brussels, Belgium.
- [42] **2018**. *Adjunction to movement paths: Floating quantifiers as the little brother of parasitic gaps*. Poster presented at the 49th Annual Meeting of the North East Linguistic Society (NELS 49), October 5–7, Cornell University, Ithaca, NY.
- [41] **2018** (with Alëna Aksënova^{SBU}, Hyunah Baek^{SBU}, Aniello De Santo^{SBU}, Hossep Dolatian^{SBU}, Sedigheh Moradi^{SBU}, Jon Rawski^{SBU}, Suji Yang^{SBU}, and Jeffrey Heinz). *Tiers and relativized locality across language modules*. Slides of a talk given at the 1-day workshop *Parallels Between Phonology and Syntax*, July 9, Meertens Instituut, Amsterdam, Netherlands.
- [40] **2018**. *Grammar size and quantitative restrictions on movement*. Slides of a talk given at *SCiL 2018*, January 4–7, Utah.
- [39] **2017**. *Locality domains and phonological c-command over strings*. Slides of a talk given at the 48th Annual Meeting of the North East Linguistic Society (NELS 48), October 27–29, University of Iceland, Reykjavík, Iceland.
- [38] **2017**. *Syntax in phonology? C-command over strings*. Slides of a talk given at *NECPhon 2017*, October 21, Stony Brook University, Stony Brook, NY.
- [37] **2017** (with Aniello De Santo^{SBU} and John E. Drury). *Evaluating subregular distinctions in the complexity of generalized quantifiers*. Poster presented at the *ESSLLI Workshop on Quantifiers and Determiners (QUAD 2017)*, July 17 – 21, University of Toulouse, France.
- [36] **2017**. *Graph transductions and typological gaps in morphological paradigms*. Slides of a talk given at the 15th meeting on the mathematics of language (*MOL 2017*), July 13–14, Queen Mary University of London, London, UK.
- [35] **2016** (with Alëna Aksënova^{SBU} and Sedigheh Moradi^{SBU}). *Tier-based strict locality in phonology and morphology*. Slides of a talk given at the 10th Northeast Computational Phonology Meeting (*NECPhon*), September 24, UMass Amherst, Amherst, MA.
- [34] **2016** (with Alëna Aksënova^{SBU} and Aniello De Santo^{SBU}). *A single movement normal form for Minimalist grammars*. Slides of a talk given at the 21st Conference on Formal Grammar, August 20–21, Bolzano, Italy.
- [33] **2016** (with Alëna Aksënova^{SBU} and Sedigheh Moradi^{SBU}). *Morphotactics as tier-based strictly local dependencies*. Slides of a talk given at the 14th *SIGMORPHON Workshop on Computational Research in Phonetics, Phonology, and Morphology*, August 11, Berlin, Germany.

- [32] **2016** (with Aniello De Santo^{SBU} and Alëna Aksënova^{SBU}). *An alternate view on strong lexicalization in TAG*. Slides of a talk given at the 12th International Workshop on Tree Adjoining Grammars and Related Formalisms, June 29 – July 1, Düsseldorf, Germany.
- [31] **2015** (with Brigitta Fodor^{SBU}, James Monette^{SBU}, Gianpaul Rachiele^{SBU}, Aunika Warren^{SBU}, and Chong Zhang^{SBU}). *A refined notion of memory usage for Minimalist parsing*. Slides of a talk given at MOL14, July 25–26, University of Chicago, Chicago, IL.
- [30] **2015**. *A hidden consensus: Computational invariants of Minimalist syntax*. Poster presented at The Road Ahead, May 28–30, Athens, Greece.
- [29] **2015** (with Jeffrey Heinz). *Commonality in disparity: The computational view of syntax and phonology*. Slides of a talk given at GLOW 2015, April 18, Paris, France.
- [28] **2015**. *A computational guide to the dichotomy of features and constraints*. Slides of a talk given at DGfS 2015, April 15–18, University of Leipzig, Leipzig, Germany.
- [27] **2014**. *Dependencies in syntax and phonology: A computational comparison*. Slides of a talk given at NECPHON 2014, November 15, NYU, New York, New York.
- [26] **2014**. *Models of adjunction in Minimalist grammars*. Slides of a talk given at FG2014, August 16–17, University of Tübingen, Tübingen, Germany.
- [25] **2014** (with Bradley Marcinek^{SBU}). *Evaluating evaluation metrics for Minimalist parsing*. Slides for a talk given at Cognitive Modeling and Computational Linguistics (CMCL 2014), June 26, Baltimore, MD.
- [24] **2014**. *Late Merge as lowering movement in Minimalist grammars*. Slides of a talk given at LACL 2014, June 18–20, IRIT, Toulouse, France.
- [23] **2014**. *Feature geometry and the person case constraint: An algebraic link*. Slides of a talk given at CLS 50, April 10–12, University of Chicago, Chicago, IL.
- [22] **2013**. *The syntactic algebra of adjuncts*. Slides of a talk given at CLS 49, April 18–22, University of Chicago, Chicago, IL.
- [21] **2013**. *The price of freedom: Why adjuncts are islands*. Slides of a talk given at the Deutsche Gesellschaft für Sprachwissenschaft 2013, March 12–15, University of Potsdam, Potsdam, Germany.
- [20] **2013**. *Of tops and bottoms: The algebra of person case constraints*. Slides of a talk given at the Southern California Students in Linguistics Conference 2013 (SCSiL 2013), January 18, University of Southern California, Los Angeles, CA.
- [19] **2012** (with Natasha Abner). *Is syntactic binding rational?* Poster presented at the 11th International Workshop on Tree Adjoining Grammars and Related Formalisms (TAG+11), September 26–28, University Paris-Diderot, Paris, France.
- [18] **2012**. *Tree adjunction as Minimalist lowering*. Slides of a talk given at the 11th International Workshop on Tree Adjoining Grammars and Related Formalisms (TAG+11), September 26–28, University Paris-Diderot, Paris, France.
- [17] **2012**. *Movement-generalized Minimalist grammars*. Slides of a talk given at LACL 2012, July 2–4, LINA, Nantes, France.

- [16] **2012** (with Natasha Abner). *Binding complexity and the status of pronouns in english and asl*. Poster presented at Formal and Experimental Advances in Sign Language (FEAST), June 1–2, University of Warsaw, Warsaw, Poland.
- [15] **2012** (with Natasha Abner). *The complexity of binding in English and ASL*. Slides of a talk given at CLS 48, April 19–21, University of Chicago, Chicago, IL.
- [14] **2012**. *Why there must be exceptions to Principle B*. Poster presented at the 30th West Coast Conference on Formal Linguistics (WCCFL 30), April 13–15, UCSC, Santa Cruz, CA.
- [13] **2011**. *Locality and the complexity of Minimalist derivation tree languages*. Slides of a talk presented at the 16th Conference on Formal Grammar (FG-2011), August 6, 2011, Ljubljana, Slovenia.
- [12] **2011**. *Closure properties of Minimalist derivation tree languages*. Slides of a talk given at LACL 2011, June 29–July 1, LIRMM, Montpellier, France.
- [11] **2011**. *Optimality is not a race: Against a performance-based view of reference-set computation*. Slides of a talk given at Glow 34, April 28–30, 2011, University of Vienna, Vienna, Austria.
- [10] **2011**. *Lost in translation: A formal model of Merge-over-Move and its implications for the language faculty*. Slides of a talk given at ConSOLE XIX, January 5–8, 2011, University of Groningen, Groningen, Netherlands.
- [9] **2010**. *Efficient computation at the interfaces*. Poster presented at On Linguistic Interfaces II (OnLI II), December 2–4, 2010, University of Ulster, Belfast, United Kingdom.
- [8] **2010**. *Rethinking transderivationality*. Slides of a talk given at the 41st annual meeting of the North East Linguistic Society (NELS 41), October 22–24, University of Pennsylvania, Philadelphia, PA.
- [7] **2010**. *Reference-set constraints as linear tree transductions via controlled optimality systems*. Slides of a talk presented at the 15th Conference on Formal Grammar (FG-2010), August 7–8, 2010, University of Copenhagen, Copenhagen, Denmark.
- [6] **2010**. *Concealed reference-set computation or how syntax escapes the parser's clutches*. Poster presented at The Language Design, May 27–29, UQAM, Montreal, Canada.
- [5] **2009**. *Locality in flux — reducibility results for syntactic constraints*. Poster presented at the Conference on Minimalist Approaches to Syntactic Locality (MASL), August 26 – 28, Research Institute for Linguistics of the Hungarian Academy of Sciences, Budapest, Hungary.
- [4] **2009**. *Some interdefinability results for syntactic constraint classes*. Slides of a talk given at the Eleventh Meeting on the Mathematics of Language (MOL11), August 20 – 21, Universität Bielefeld, Bielefeld, Germany.
- [3] **2009**. *Towards a factorization of string-based phonology*. Slides of a talk given at the 14th ESSLI Student Session, July 20 – 31, Université Bordeaux, Bordeaux, France.
- [2] **2009**. *Comparing incomparable frameworks — a model theoretic approach to phonology*. Hand-out of a talk given at the 33rd Penn Linguistics Colloquium (PLC33), March 27 – 29, University of Pennsylvania, Philadelphia, PA.

- [1] **2008.** *Ein modelltheoretischer Blick auf syntaktische Constraints*. Handout of a talk given at Generative Grammatik des Südens 2008 (GGS 08), May 2-4, ZAS Berlin, Germany.

Awards, Grants & Prizes

2019–2025	NSF CAREER Award Project: <i>Abstract Universals of (Morpho)Syntax: Computational Characterizations and Empirical Implications</i> National Science Foundation
2023–2025	OVPR Seed Grant Program (as Co-PI; PI Tom McFadden, Co-PI John Bailyn, Co-PI Sandhya Sundaresan) Project: <i>Syntactic Locality: An Interface of Typology, Theory and Computation</i> Stony Brook University
2025	FAHSS Grant for Interdisciplinary Projects (as PI; Co-PIs Gary Mar and Jeffrey Heinz) Project: <i>Conference "Mathematics of Language (MOL) 2025"</i> Stony Brook University
2022	Dean's Award for Excellence in Graduate Mentoring by a Faculty Member Stony Brook University
2020–2021	IACS Seed Grant Project: <i>Parsing Protein Chains: Leveraging Computational Linguistics for Faster Protein Folding</i> Stony Brook University
2019	Dean's Award for Excellence in Graduate Teaching by a Faculty Member Stony Brook University
2018	Grant for Teaching and Learning with New Technology (TALENT) Project: <i>Using Jupyter Notebooks as an Interactive, Open Source Learning Platform for Mathematics in Linguistics</i> Stony Brook University President's Distinguished Travel Grant Award Stony Brook University AHLSS Graduate Fellows and Faculty Research Program Stony Brook University
2015	Research Initiative Grant for Faculty in the Fine Arts, Humanities, and Lettered Social Sciences (FAHSS) Project: <i>Human Sentence Processing as a Window into Efficient Structure Inference</i> Stony Brook University
2014	E.W. Beth Dissertation Prize for Outstanding Dissertations in Logic, Language, and Information Association for Logic, Language and Information
2010–2011	DOC-Fellowship of the Austrian Academy of Sciences Austrian Academy of Sciences (ÖAW)

2009 Springer Prize for Best Student Paper at ESSLLI2009
ESSLLI Student Session

Advisees (co-advised students are marked with *)

Matthew Hayden expected date of graduation May 2029

Logan Swanson* expected date of graduation May 2027

Kenneth Hanson Ph.D. July 2025
 “The subregular complexity of case and agreement”
 Placement upon graduation: Lecturer, Yale University

Nazila Shafiei Ph.D. March 2022
 “Computational locality and domain of syntactic long-distance dependencies”
 Placement upon graduation: Natural Language Engineer, Amazon

Lei Liu* Ph.D. January 2022
 “Phrasal weight effect on word order”
 Placement upon graduation: Post-doc, Department of Linguistics, University of Leipzig, Germany

Sedigheh Moradi* Ph.D. August 2021
 “Monotonicity in morphosyntax”
 Placement upon graduation: Natural Language Engineer, Amazon

Alëna Aksënova Ph.D. May 2020
 “Tool-assisted induction of subregular languages and mappings”
 Placement upon graduation: Natural Language Engineer, Google

Aniello De Santo Ph.D. May 2020
 “Structure and memory: A computational model of storage, gradience, and priming”
 Placement upon graduation: Assistant Professor, Department of Linguistics, University of Utah

Chong Zhang Ph.D. August 2017
 “Stacked relatives: their structure, processing and computation”
 Placement upon graduation: Natural Language Engineer, Kasisto

Teaching (new courses developed by me are marked with *)

Stony Brook LIN 120 Language and Technology*
 LIN 311 Syntax
 LIN 335 Computational Linguistics*
 LIN 346 Language and Meaning
 LIN 361/539 Mathematical Methods in Linguistics*
 LIN 538 Statistics for Linguists
 LIN 626 Computational Phonology
 LIN 628 Computational Syntax*

LIN 630 Parsing and Syntactic Processing*

LIN 637 Computational Linguistics 2*

LIN 651 Syntax seminar

Department Service

Leadership Positions

Department Chair	since Fall 2026
Ph.D. Program Director	Fall 2021 – Spring 2024
M.A. Linguistics Program Director	Fall 2021 – Spring 2026
M.A. CompLing Program Director	Fall 2017 – Spring 2024

Program Development

M.A. Computational Linguistics	2014 & 2015, program active since Fall 2017
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Course Development

LIN 120 Language & Technology
LIN 335 Computational Linguistics
LIN 361/539 Mathematical Methods in Linguistics
LIN 628 Computational Syntax
LIN 630 Parsing and Syntactic Processing
LIN 637 Computational Linguistics 2

Admissions Committees

M.A. Computational Linguistics	2024, 2023, 2022, 2021, 2020, 2019, 2018, 2017
M.A. Linguistics	2025, 2024, 2023, 2022, 2021
Ph.D. Linguistics	2023, 2022, 2021, 2017, 2016

Search Committees

2023	Assistant to the Chair (ATC) Visiting Assistant Professor in Phonology
2020	Assistant Professor Computational Linguistics
2017	Associate Professor Computational Linguistics

Other Committees

Awards Committee	2024
Colloquium Committee	2019, 2018, 2017, 2016, 2015, 2014
Merit Raise Committee	2025
Strategic Plan Committee	2021, 2015
Student Research Funds Committee	2024, 2023, 2022, 2021, 2016

Other activities

Co-director of CompLab	2017, 2016, 2015, 2014, 2013
MathLing Reading Group organizer	2024, 2023, 2022, 2020, 2019, 2018, 2017, 2016, 2015, 2014
Open House organizer	2024, 2023, 2022
Run Run Shaw lecture organizer	Matilde Marcolli (joint with Mathematics), 2024

University Service

CAPRA (Academic Planning & Resource Allocation)	2025, 2024, 2023
Mathematics Summer Program	2024, 2023, 2022, 2019, 2018, 2017, 2016
OVPR Seed Grant Reviewer	2024
Stony Brook NSF CAREER Q&A Panel	2023, 2022, 2020
Provostial Working Group on Graduate Education	2022
IACS Computes! High School Summer Camp	2018, 2017
STEM SBC Committee	2017
Stony Brook Visit a Class Program	2017, 2016
Stony Brook Career Fair	2016

Service to the Field (see also the reviewing tables at the end of the CV)

Paper reviewing	ACL • Biolinguistics • Biology • Catalan Journal of Linguistics • Cognitive Modelling & Computational Linguistics • ESSLLI Student Session • ExpSem • Formal Grammar • Glossa • ICLR • Italian Journal of Linguistics • Journal of Logic, Language, and Information • Journal of Language Modelling • Journal of Linguistics • Languages • Linguistic Inquiry • Mathematics of Language • NAACL-HLT • Natural Language & Computer Science • Natural Language & Linguistic Theory • Phonology • Routledge • SCiL • STAR • Syntax • TAG+
Book reviewing	Cambridge University Press • Oxford University Press
Abstract reviewing	AAAI-SSS • AIMM • AMP • Austrian Linguistics Conference (OELT) • Empirical Issues in Syntax and Semantics (EISS) • FASL • LabPhon • PLC • Syntax and Semantics in Paris (CSSP) • WCCFL • XPRAG
2024–present	Vice-President and President Elect of the ACL Special Interest Group for the Mathematics of Language (SIGMOL)
2024–present	Editorial board member of the <i>Journal of Language Modelling</i>
2023, 2024	NSF panelist
2023–present	Organizer of the MG+ online workshop series on Minimalist grammars
2019–present	Creator and main contributor of the blog Outdex

2009–2013 Creator of the *uclawpl* $\text{\LaTeX}$ 2 ϵ class file used by the **UCLA Working Papers
in Linguistics**

Languages

Natural	in decreasing order of proficiency: German, English, French, Icelandic, Old Norse, Latin, Spanish
Markup	CSS, HTML, $\text{\LaTeX}$, Markdown, Tikz
Programming	Awk, Bash, Lua, Python

Paper reviewing

Paper Reviews in 20...	≤ 13	14	15	16	17	18	19	20	21	22	23	24	25
ACL								✓					
Biolinguistics			✓	✓									
Biology									✓				
Cambridge University Press	✓										✓		✓
Catalan Journal of Linguistics						✓							
Cognitive Modelling & Computational Linguistics			✓	✓	✓	✓	✓						
ESSLLI Student Session	✓												
ExpSem								✓	✓				
Formal Grammar					✓	✓	✓	✓					
Glossa					✓	✓		✓	✓	✓			
ICLR								✓					
Italian Journal of Linguistics		✓											
Journal of Logic, Language, and Information					✓					✓			
Journal of Language Modelling				✓			✓	✓		✓		✓	✓
Journal of Linguistics			✓										
Languages											✓		
Linguistic Inquiry	✓	✓					✓	✓					
Mathematics of Language							✓						
NAACL-HLT						✓							
Natural Language & Computer Science						✓							
Natural Language & Linguistic Theory	✓												
Phonology					✓	✓	✓						
Oxford University Press	✓		✓										
Routledge							✓						
SCiL						✓	✓	✓		✓	✓	✓	✓
STAR (Syntactic Theory and Research)												✓	
Syntax					✓			✓	✓	✓			
TAG+					✓								

Abstract reviewing

Abstract Reviews in 20...	≤ 13	14	15	16	17	18	19	20	21	22	23	24	25
AAAI-SSS					✓								
AIMM							✓						
AMP										✓	✓	✓	
Austrian Linguistics Conference (OELT)									✓				
Empirical Issues in Syntax and Semantics (EISS)										✓			
FASL							✓						
LabPhon				✓									
Penn Linguistics Colloquium (PLC)											✓	✓	✓
Syntax and Semantics in Paris (CSSP)									✓				
WCCFL		✓	✓		✓	✓		✓					
XPrag					✓								